

HURON, SD, USA

WMO: 726540

Lat: 44.383N

Lon: 98.217W

Elev: 393

StdP: 96.69

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14936

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.2	-22.3	-28.7	0.3	-24.9	-25.8	0.4	-22.0	15.6	-11.5	14.0	-9.3	3.9	320	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	34.1	23.3	32.2	22.7	30.6	21.8	25.2	31.0	24.2	30.0	23.2	28.8	5.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.3	28.9	22.4	17.9	27.8	21.3	16.7	26.9	79.5	31.0	75.1	30.1	71.0	28.8	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.6	11.1	9.6	-30.2	37.7	4.1	2.4	-33.1	39.5	-35.5	40.9	-37.9	42.3	-40.9	44.0		
			-30.3	26.8	4.0	1.0	-33.2	27.5	-35.5	28.1	-37.8	28.7	-40.7	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-9.2	-6.5	0.4	7.7	14.5	20.2	23.5	22.0	17.4	9.1	0.9	-6.3
	DBStd	12.69	7.36	7.57	7.49	5.76	4.83	3.83	3.55	3.34	4.95	5.42	6.28	6.75
	HDD10.0	2354	596	461	307	108	14	0	0	0	4	82	278	504
	HDD18.3	4249	854	694	557	320	138	23	4	8	76	288	524	762
	CDD10.0	1582	0	0	9	39	153	307	417	371	225	55	4	0
	CDD18.3	436	0	0	1	2	18	81	163	121	48	3	0	0
	CDH23.3	4636	0	0	6	43	237	838	1766	1177	516	52	1	0
	CDH26.7	1759	0	0	1	9	69	283	761	440	184	12	0	0
(14) Wind	WSAvg	4.6	4.7	4.8	5.1	5.3	4.9	4.3	4.0	3.9	4.3	4.6	4.7	4.6
(15) Precipitation	PrecAvg	543	12	18	35	59	75	91	68	57	52	43	20	14
	PrecMax	805	64	98	150	169	195	292	170	164	170	138	100	42
	PrecMin	279	1	1	4	5	8	20	5	4	0	2	0	0
	PrecStd	129	11	16	28	37	45	53	43	32	36	35	21	11
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	14.1	23.0	27.6	31.9	34.2	37.6	35.7	34.3	28.5	20.8	11.5
		MCWB	5.3	8.5	12.7	15.7	20.0	23.1	24.3	23.9	21.4	17.3	12.2	6.1
	2%	DB	5.0	9.6	18.5	23.9	28.5	31.7	34.6	32.9	31.2	24.2	16.4	7.6
		MCWB	2.1	5.2	11.5	13.9	18.4	22.4	24.1	23.0	20.9	15.8	9.9	3.5
	5%	DB	2.8	6.2	14.6	20.7	26.0	29.8	32.7	30.9	28.7	21.1	12.8	4.5
		MCWB	1.0	3.3	8.9	12.1	17.4	21.4	23.4	22.3	19.6	14.0	7.6	1.8
	10%	DB	1.0	3.4	11.0	17.8	23.5	28.0	30.8	29.1	26.2	18.0	9.8	2.2
		MCWB	-0.2	1.5	7.3	10.6	16.0	20.3	22.4	21.5	18.3	12.1	5.6	0.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	9.4	14.9	17.1	21.7	25.3	27.0	26.1	23.5	19.3	13.0	6.9
		MCDB	9.8	14.4	19.8	25.3	28.8	31.8	33.2	31.9	29.8	25.2	18.9	11.1
	2%	WB	2.5	5.5	12.2	14.8	19.9	23.9	25.7	24.7	22.3	16.7	10.3	3.9
		MCDB	4.6	9.0	17.6	22.0	26.4	30.1	31.8	30.1	29.0	21.8	15.6	7.3
	5%	WB	1.1	3.4	9.6	12.8	18.4	22.6	24.7	23.6	21.1	14.8	7.9	2.0
		MCDB	2.7	6.0	14.5	19.2	24.6	28.3	30.6	29.0	27.0	20.1	12.5	4.1
	10%	WB	-0.2	1.5	6.8	11.3	16.8	21.4	23.6	22.6	19.8	12.8	5.9	0.6
		MCDB	1.1	3.3	10.9	16.9	22.3	26.7	29.3	27.7	24.7	17.4	9.5	2.1
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	10.8	11.1	12.7	12.3	12.1	12.6	12.6	13.6	12.6	11.4	9.7
		MCDBR	12.3	13.7	17.2	17.9	16.0	13.8	14.2	14.5	16.0	17.2	16.1	13.2
	5% WB	MCWBR	10.0	9.8	10.4	9.6	8.2	6.7	6.1	6.5	7.4	9.6	10.0	9.8
		MCDWR	12.1	13.2	16.6	16.1	13.9	12.2	12.8	12.5	13.6	14.9	15.8	12.4
(40) Clear-Sky Solar Irradiance	taub		0.265	0.290	0.312	0.360	0.391	0.381	0.391	0.393	0.348	0.316	0.285	0.266
		taud	2.450	2.395	2.431	2.307	2.298	2.382	2.364	2.341	2.446	2.500	2.494	2.458
	Ebn at Noon		873	904	928	901	879	888	873	858	877	865	841	835
		Edh at Noon	70	89	99	122	128	118	119	117	97	80	67	63
(44) All-Sky Solar Radiation	RadAvg		1.84	2.88	3.84	4.87	5.74	6.44	6.78	5.63	4.45	2.94	1.98	1.44
	RadStd		0.16	0.25	0.36	0.42	0.47	0.45	0.20	0.45	0.30	0.36	0.20	0.12

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page